Interactive Simulation of Dexterous Manipulation with Hand Tracking: Robot Hand Bolting and Multi-User Collaborative Simulation in VR

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Motivation

- **Hand tracking for dexterous manipulation**
  - Hand tracking is essential in teleoperation, learning from demonstration (LfD), AR/VR applications, and education/training purposes
  - Accurate and robust tracking is essential to capture intricate and rapid human hand motions



Fig 1. Tesla (Learning from Demonstration)



Fig 2. Meta (AR/VR)

Contribution

- We demonstrate **two interactive dexterous manipulation simulations using hand tracking**:

  1) Robot hand bolting; and **2) Multi-user hand simulation**
  - **Robust, fast, and accurate hand tracking** with Visual-Inertial Skeleton Tracking (VIST)
  - **Real-time and accurate simulation** for contact intensive tasks and realistic haptic feedback

Hand Tracking + Haptic System

- **Visual-inertial Skeleton Tracking (VIST)**[1]
  - Complementary use of vision and inertial information
  - **Precise (i.e., tracking RMSE 8.4mm)** and **fast (i.e., 292Hz)** tracking
  - **Robust** to visual occlusion, electromagnetic interference, and contact

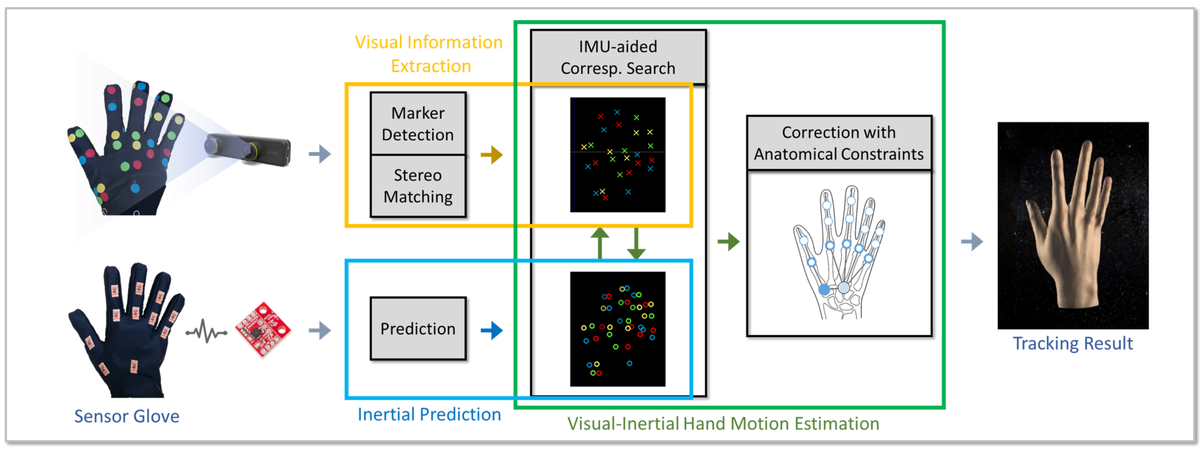


Fig 3. Visual-inertial fusion pipeline

- **Hand tracking results**

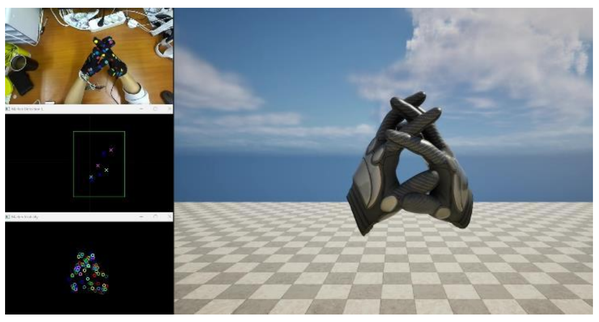


Fig 4. Intertwined fingers



Fig 5. Overlapping hands



Fig 6. Various hand motions

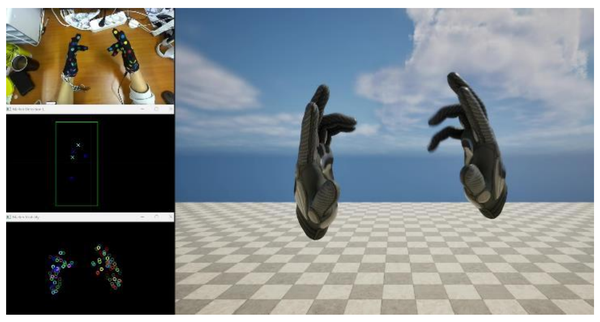


Fig 7. Visually occluded fingers

- **Haptic feedback device**
  - **Cutaneous haptic device (CHD)**[2]: underactuated cable-driven parallel mechanism, generating 3 DoF contact force on the fingertip by using hybrid orientation-force control
  - **Vibrotactile modules** on palmar side of the fingers

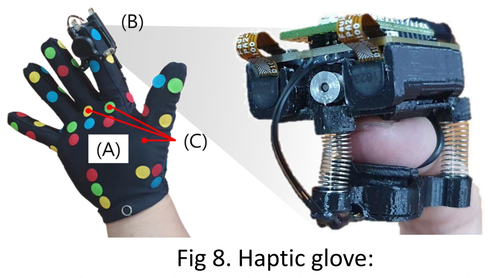


Fig 8. Haptic glove:
(A) VIST; (B) CHD; (C) vibrotactile modules

Demonstration 1: Robot hand bolting

- **Challenges**
  - Control of a high-degree-of-freedom robot hand to achieve tight-tolerance bolting task
  - Accurate simulation involving non-convex and complex geometries
- **Key features**
  - **Teleoperation of a robot hand in VR** using VIST hand tracking
  - **Bolting simulation** with our **multi-contact high-precision robot simulator**[3] that utilizes neural network-based collision detection and data-driven contact point clustering method
  - **Haptic feedback (CHD + vibrotactile)** to enhance immersion

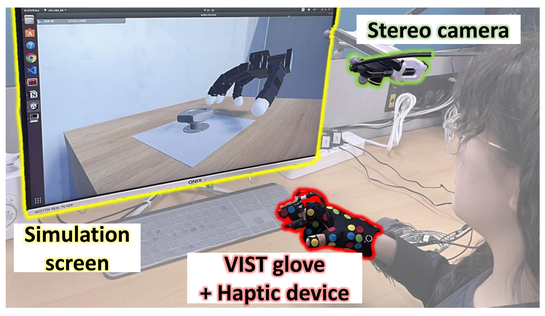


Fig 9. Robot hand bolting setup

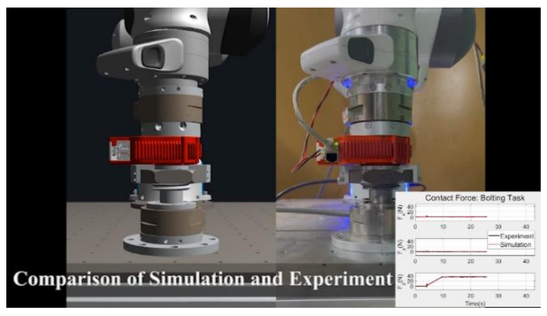


Fig 10. Tight-tolerance bolting simulation



Fig 11. Robot hand bolting simulation

Demonstration 2: Multi-user hand simulation

- **Multi-user hand simulation system**
  - **Collaborative task of multiple users in VR**
  - Server computer for simulation and user computers for user interface
  - **Fast (GPU parallelization, 120Hz) and accurate simulation** based on **subsystem-based ADMM framework**[4] in the server computer given the hand motions from user computers
  - Each user computer tracks the hand motion and provides the visual/haptic feedback given the simulation information (i.e., object states and contact forces) from the server

(A) VIST glove + Haptic device
(B) Stereo camera
(C) Simulation screen

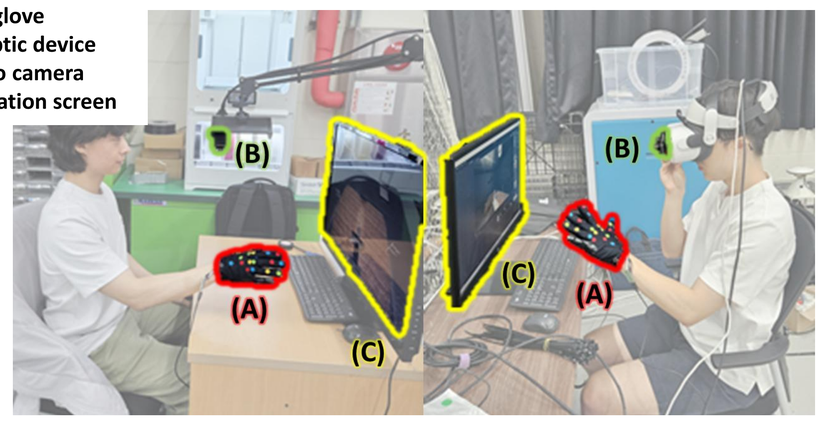


Fig 12. Multi-user hand simulation setup



Fig 13. Collaborative block stacking task: user #1 hand (top-left), and user #2 hand (bottom-right)

Acknowledgement

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[1] Y. Lee, W. Do, H. Yoon, J. Heo, W. Lee and D. J. Lee, "Visual-inertial hand motion tracking with robustness against occlusion, interference, and contact," Science Robotics, 2021
[2] S. Lee, H. Kim and D. J. Lee, "Symmetry-Based Modeling and Hybrid Orientation-Force Control of Wearable Cutaneous Haptic Device," IROS, 2023
[3] J. Yoon, M. Lee, D. Son and D. J. Lee, "Fast and Accurate Data-Driven Simulation Framework for Contact-Intensive Tight-Tolerance Robotic Assembly Tasks," arXiv, 2022
[4] J. Lee, M. Lee and D. J. Lee, "Modular and Parallelizable Multibody Physics Simulation via Subsystem-Based ADMM," ICRA, 2023